

PARAS BANSAL

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Education

Indian Institute of Information Technology, Allahabad <i>B.Tech in Information Technology; Minor: Science of Happiness; CGPA: 9.00 / 10</i>	Aug 2024 – May 2028 <i>Prayagraj, Uttar Pradesh</i>
Golden Earth Global School, Sangrur <i>Class 12 (CBSE); Percentage: 94.6%</i>	Apr 2023 – Mar 2024 <i>Sangrur, Punjab</i>
Golden Earth Global School, Sangrur <i>Class 10 (CBSE); Percentage: 97%</i>	Apr 2021 – Mar 2022 <i>Sangrur, Punjab</i>

Key Projects

Pseudo2CPP C++, NLP, Compiler Design, DSA GitHub - Developed an NLP-based compiler/transpiler in C++ converting English algorithmic queries into optimized DSA implementations using 50+ reusable templates. - Designed a semantic keyword engine analyzing 100+ tokens to classify algorithms with 90%+ accuracy. - Built modular parsing, semantic analysis, and code generation pipelines enabling automated competitive programming boilerplate with O(1) template retrieval.	Feb 2026 – Mar 2026
Energy-Aware Beam-ACO Routing C++, Metaheuristics, Optimization GitHub - Implemented a hybrid Ant Colony Optimization (ACO) + Beam Search routing algorithm in C++, reducing path-planning cost by 15%. - Integrated Beam Search top-K pruning to limit exploration space, improving convergence speed by 40%. - Designed a physics-based energy cost model using velocity and drag, improving routing efficiency by 15%.	Oct 2025 – Dec 2025
Samadhaan MERN Stack, REST APIs, MongoDB GitHub Live Server - Developed a scalable MERN-based grievance management platform supporting 12+ civic complaint categories with real-time status tracking. - Built and documented 15+ RESTful APIs using Node.js, Express.js with JWT authentication and role-based access control. - Optimized backend performance using MongoDB indexing, reducing API latency by 20% and improving UI load time by 25%.	Jun 2025 – Aug 2025

Technical Skills

Languages: C++, C, Java, Python, JavaScript (ES6+), SQL
Web: HTML5, CSS3, React.js, Node.js, Express.js, Tailwind CSS
Databases: MongoDB, MySQL
Tools & Platforms: Git, GitHub, Linux
Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Theory of Computation, Computer Architecture

Competitive Programming

Codeforces (**Expert**, Rating 1710) | CodeChef (**4-Star**, Rating 1820) | LeetCode (**Knight**, Rating 1914)

Achievements

- JEE Main 2024: AIR 3341** (Top 0.2% among 1.4M candidates); JEE Advanced 2024: **AIR 9232**.
- Codeforces Expert** (Top 5% globally) with rating **1710**; ranked **830 / 91,904+** active users in India with best contest rank **451**.
- Secured **Global Rank 38** in LeetCode Weekly Contest 484.
- Achieved **Global Rank 93** in CodeChef Starters 224.
- Solved **500+ Data Structures & Algorithms problems** across Codeforces, CodeChef, and LeetCode.

Leadership & Volunteering

Head Boy <i>Golden Earth Global School</i> Led a student council of 35 members to organize 10+ school events, improving school-wide communication.	Class 10
Student Mentor <i>Prayaas (Social Service)</i> Mentored 100+ underprivileged students; developed resources rated 90% Excellent.	2024 – Present